

P-204A Boiler Feed Water Pump NDT & Vibration Survey

Confidential Maintenance Inspection • Boiler & Utilities Section

1. Asset Overview & Service History

Equipment Tag: P-204A Multi-Stage Boiler Feed Water Pump. Location: Crude Distillation Unit (CDU).
Design Flow Rate: 320 m3/hr. Total Dynamic Head: 850 meters. Operating Temperature: 165°C.
Last Overhaul: 14 months prior. Cumulative Run Hours: 12,480 hrs.

2. Inspection Findings & Cavitation Diagnostics

Field operators reported audible rattling and cracking sound resembling gravel passing through first-stage impeller casing.
High-frequency acoustic emission testing confirmed active hydraulic cavitation during minimum-flow recirculation operations.
Suction strainer differential pressure reached 0.85 bar (normal limit: 0.25 bar), causing severe Net Positive Suction Head Available (NPSHa) deficit.
Thrust bearing axial displacement sensor recorded +0.32 mm excursion toward drive end.

3. Visual Casing & Seal Examination

External visual inspection revealed localized crystallization and dried condensate seepage around mechanical seal gland packing.
Thermographic imaging indicated hot spot on outboard thrust bearing housing reaching 84.5°C (Alarm threshold: 80°C).
Coupling disc pack showed slight fretting corrosion on intermediate spacer bolts.

4. Engineering Recommendations

- Clean suction duplex strainer immediately during off-peak window to restore NPSHa.
- Switch operating duty to standby pump P-204B.
- Conduct ultrasonic casing thickness inspection on suction spool to check for cavitation erosion.
- Drain and replenish ISO VG 46 synthetic turbine lube oil.